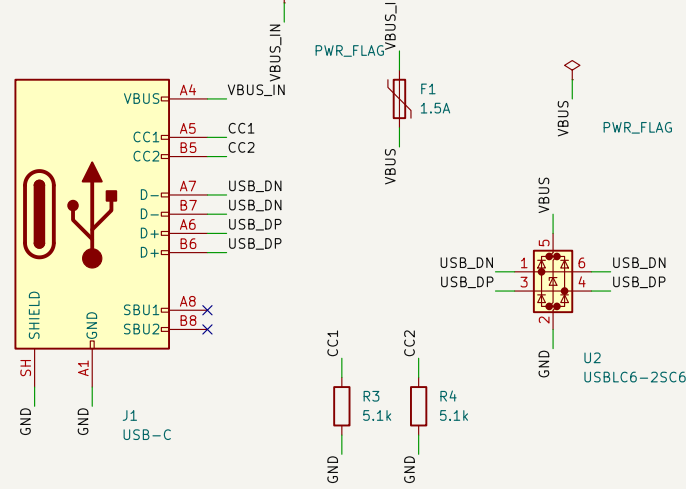
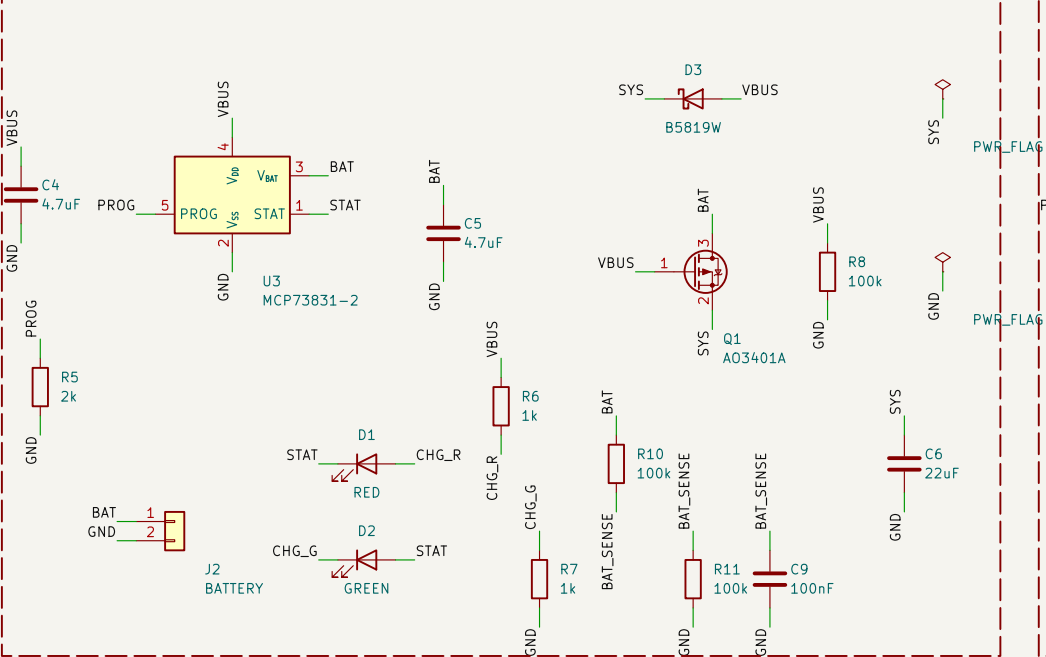


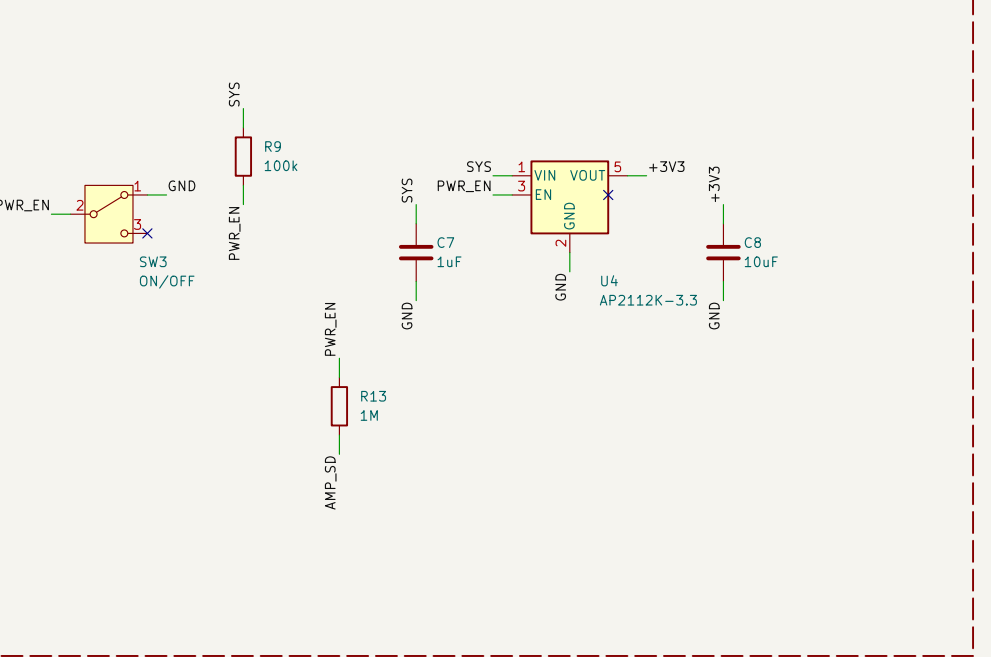
USB-C input and protection



Charger, power path and battery

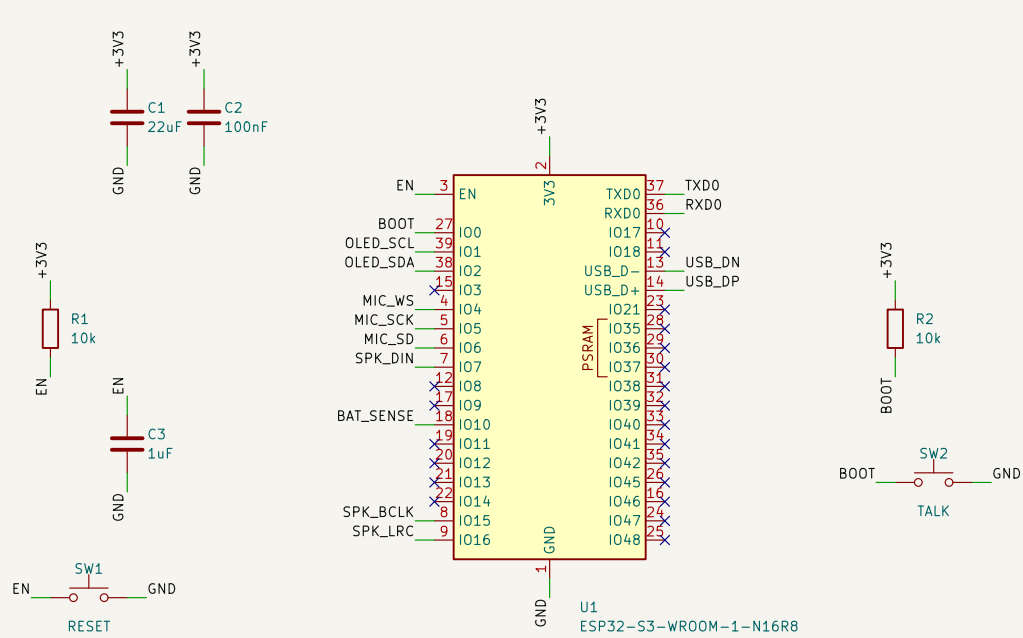


On/off switch and 3.3 V regulator

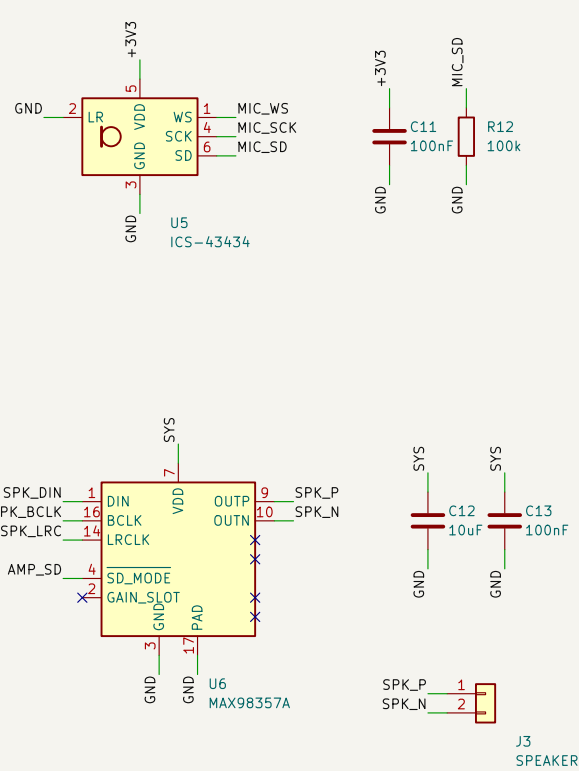


GPIO map = firmware/nandi/config.h. Talk button = GPIO0 (BOOT). Battery sense = GPIO10 (2:1).

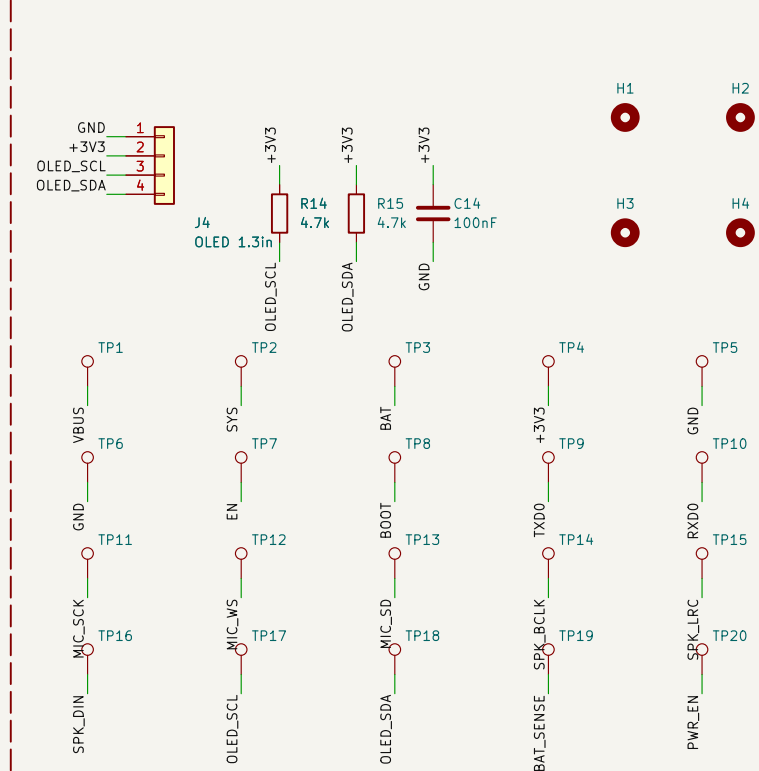
ESP32-S3 module, reset and talk button



Microphone (I2S in) and amplifier (I2S out)



OLED, test points, mounting holes



Flash over USB-C (native USB). Arduino IDE: USB CDC On Boot = Enabled. Speaker: 4 or 8 ohm.

Battery: 1-cell Li-ion/LiPo WITH its own protection board. Check the plug polarity: pin 1 = +.

MAX98357A SD_MODE via 1M from PWR_EN = (L+R)/2 mono; GAIN_SLOT open = 9 dB. Charge current 500 mA (R5 = 2k).

GPIO map matches firmware/nandi/config.h (FW 2.1.0)		
ESP32-S3 voice assistant: I2S mic, I2S amp, OLED, Li-ion charging, USB-C		
Team Novadian, IERT Prayagraj		
Sheet: /		
File: nandi-pcb.kicad_sch		
Title: Nandi v1 main board		
Size: A3	Date: 2026-09-21	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 1/1